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Assignment 12

Final Proposal

Due Date: June 23th, 2026.

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Problem Statement

Emergency departments in resource-constrained Caribbean settings like Mercer General Hospital operate on paper-based triage systems that generate no structured, queryable data. Without a digital foundation, locally validated AI triage tools cannot be built, and globally trained models — developed predominantly on North American and European patient populations — cannot be reliably applied. This project proposes to establish a structured digital triage dataset from the Mercer General ED as the necessary first step toward a clinically credible, regionally relevant AI-assisted triage classifier.

Project Context

This project is structured as two phases, referenced repeatedly throughout the proposal, diagrams, and weekly write-ups. They are not interchangeable — each has a different scope, timeline, and deliverable.

Phase 1 — the 12-week pilot (this programme). The goal is to design and validate a structured **digital triage capture tool** that mirrors the existing Mercer General paper triage form. The output is not an AI model — it is a clean, de-identified, queryable dataset built from real ED encounters. Mercer currently runs entirely on paper, which means no structured triage data exists anywhere to train or validate a model against. Phase 1 exists to fix that gap first.

Phase 2 — beyond this programme (future work). Once a sufficient digital dataset exists from Phase 1, Phase 2 would use that data to train and locally validate a **rule-based ESI classifier**, iterating toward a machine learning model as data volume grows over time. This is the long-term vision the project is building toward — but it is explicitly *not* a deliverable of the current 12-week pilot.

Why the split matters: Most AI-assisted triage literature assumes infrastructure — electronic health records, stable data pipelines — that Mercer does not have. Proposing an AI classifier without first proposing the data foundation it depends on would be building on top of nothing. Phase 1 is the foundation. Phase 2 is the building.

Previous Works & How we Move Forward

Research into AI and machine learning in ED triage has grown significantly, with studies showing promise in earlier diagnosis and intervention, though concerns remain around overconfident outputs presenting risks to patients (Tyler et al., 2024). AI-driven triage systems have demonstrated potential by automating patient prioritisation through real-time analysis of vital signs, medical history, and presenting symptoms, with narrative reviews covering literature from 2015 to 2024 identifying both benefits and persistent limitations (Da'Costa et al., 2025).

A systematic review of prospective studies found that while AI exhibits potential in emergency patient triage, it has not yet established evidence for substituting human involvement — a critical distinction for a setting like Mercer, where nurse-led triage is the primary model and clinician trust is a known adoption barrier (Yi et al., 2025).

Infrastructure constraints, including unreliable power supply, limited internet connectivity, and the absence of EHR integration tools, alongside a lack of high-quality annotated datasets for specific populations, represent key barriers to AI triage implementation in resource-limited settings. These barriers are directly applicable to the Caribbean context. A recent scoping review examining AI triage across low- and middle-income countries found that 72% of studies were conducted in high-income countries, with LMICs representing only 28% of the evidence base — reflecting both the thinness of the evidence base in resource-constrained settings and the near-complete absence of Caribbean data from this literature (Aifuobhokhan et al., 2025).

Araouchi & Adda (2024) demonstrated a multimodal AI triage system integrating structured vitals with unstructured clinical text, showing accuracy improvements over single-modality approaches — but validated exclusively on datasets from high-income settings with complete electronic health records, a precondition Mercer does not currently meet.

Where machine learning models have been developed and internally validated in non-Western settings, results are promising but context-dependent. Sitthiprawiat et al. (2025) developed an XGBoost model across 163,452 ED visits at a Thai hospital, achieving superior predictive performance over the Canadian Triage and Acuity Scale (AUROC 0.917 vs. 0.882) — but noted that generalisability to different patient populations and healthcare infrastructures remains unproven, underscoring the need for locally trained models in every distinct clinical context.

Critically, there is precedent for the Phase 1 approach proposed here. Mitchell et al. (2024) documented triage implementation at Vila Central Hospital in Vanuatu — a Pacific Small Island Developing State with infrastructure constraints directly analogous to the Caribbean — demonstrating the feasibility of a low-cost electronic data registry system built to capture triage data in a previously paper-based ED. Their key finding was that a bespoke needs assessment of local triage requirements was essential before any digital tool could be designed, a lesson directly applicable to the Mercer context.

De Freitas et al. (2020) conducted a qualitative exploration of patient flow in a Caribbean emergency department, directly documenting the operational realities — overcrowding, delays, and resource constraints — that any digital intervention at Mercer must be designed around. This paper anchors the workflow assumptions made in the systems-thinking analysis below in real Caribbean clinical observation rather than inference from non-Caribbean settings.

New This Week: Deployment & Constraint-Focused Literature

This week's literature additions shift focus from triage accuracy toward deployment reality — the gap between a model performing well in a paper and a model surviving contact with an actual clinical workflow.

Poon et al. (2025) surveyed 43 US health systems on AI adoption during the early generative AI era and identified six recurring barriers: financial concerns, regulatory uncertainty, lack of leadership support, low clinician adoption, insufficient expertise or technology, and immature AI tools. Notably, low clinician adoption was identified as a barrier independent of model accuracy — reinforcing that a technically sound tool can still fail if it does not fit how clinicians actually work.

Salwei et al. (2024) evaluated a human-factors-based clinical decision support (CDS) tool specifically within an emergency department and found that although usability testing scored highly in both experimental and real clinical settings, actual use of the tool was low. The dominant barrier identified was a lack of workflow integration — in one documented case, an EHR system upgrade silently relocated the tool within the interface, and because the change was not communicated to physicians, it significantly disrupted their workflow and further suppressed adoption. This is a direct, real-world illustration of Constraint 1 below: a tool's clinical value collapses the moment it adds friction to an already time-pressured workflow.

Chughtai and Blanchet (2017) contributed a meta-narrative review establishing systems thinking as a methodology for public health interventions, arguing that health system interventions must be understood as embedded within a web of interacting actors, incentives, and constraints rather than as isolated technical fixes. This paper provides the methodological grounding for the workflow-mapping and constraint-identification approach taken in this proposal.

Workflow Notes

The Mercer General Emergency Department follows a door-to-disposition pathway with five measured timestamps: **arrival**, **triage**, **seen-by-physician**, **disposition decision**, and **exit**.

Stage	**Duration**	**What Happens**	**Data Captured**
Arrival	—	The patient arrives via ambulance, walk-in, or private vehicle	None yet
Registration	2–3 min	The patient's data, including their demographics, consent, and arrival modality, is collected.	Front desk — paper intake form
Triage Assessment	3–5 min	The patient's severity is assessed based on chief complaint, vitals, allergies, pain score, and ESI level assigned.	Triage nurse — paper front page
Zone Placement	—	The patient is placed in the ESI zone. Zones 1–3 equate to bed placement (resus/acute); while	None — disposition only

		zones 4–5 equate to fast-track waiting.	
Physician Assessment	ESI 2: ≤10 min ESI 3: 30–60 min	The physician accesses patient, collates their medical history, does examinations, and places orders for same.	Physician — deeper clinical pages
Investigations	30 min – 3 hr	The labs/imaging are processed, collected and accessed — **shared hospital resource.**	Lab/radiology system
Disposition Decision	At the end of the workup	The patient is either admitted/discharged/transfered based on the recommendation by the physician.	Physician
Exit	Variable	The patient leaves the hospital; however, this is delayed if admitted.	—

Key observations on information loss:

1. The triage front page (ESI, vitals, chief complaint) is **consistently completed**.
2. The deeper clinical pages (history, medication chart, disposition rationale) are **not consistently completed** — this is the first place structured data goes missing.
3. **Re-triage** depends entirely on a nurse visually scanning the waiting room at intervals. There is no structured trigger for re-assessment — a deteriorating ESI-4 patient is only caught if someone happens to notice.
4. Investigations are the dominant flow bottleneck. Lab and radiology are shared with the rest of the hospital, and ED workups compete directly with elective inpatient demand.

Workflow Diagram

In the diagram below, decision diamonds are placed at the triage-level ESI assignment step, per task instructions. Along with five annotated AI plug-in points, including one explicit **non-plug-in point** — a place where adding AI would actively make things worse.


Patient Arrival
(Ambulance/Walk-in/Private Vehicle)
Registration (2-3 min)
Demographics, Consent
1
Triage Assessment (3-5 min)
Vitals, Chief Complaint, Pain Score.
Re-triage check, 1-2 min interval based
ESI Level Assigned?
2
ESI 1
ESI 3
ESI 4-5
Resuscitation / Acute Bed
Acute Cubicle
Fast-Track Waiting Area
3
Physician Assessment
History, Exam, Orders
4
Investigations (30 min - 3 hr)
Labs / Imaging
5
Disposition Decision
Admit / Discharge /Transfer
Exit
Boarding time if admitted

AI Plug In-Key

1. **Digital Triage Capture Tool**

   Reduces the paper form with structured digital intake at Registration/Triage. This project, Phase 1.

2. **ESI Decision Support**

   Suggests an ESI level alongside the nurse's own judgement at the zone-placement decision point. This describes phase 2, future work.

3. **Re-triage Alert**

   Flags overdue waiting room re-checks based on elapsed wait time and initial vitals

4. **Structured Handover Prompt**

   Auto-populates on SBAR-style summary from digital triage data at the physician handoff.

5. **NOT a plug-in point — Investigations Bottleneck**

   AI must NOT increase test-ordering volume without addressing lab/radiology turnaround. Doing so would worsen the queue, not relieve it.

Workflow Constraints

Constraint 1 — The 3–5 Minute Triage Window (Friction Budget)

Triage nurses have a narrow assessment window and a queue of waiting patients behind every encounter. Any digital tool that asks for more fields than the existing paper form, or takes more time than handwriting, will be abandoned within days — regardless of how good the underlying model is. This directly shapes the interface design for the Phase 1 capture tool: it must replicate the paper form's speed, not add to it.

Constraint 2 — Paper-to-Digital Lag at the Point of Capture

All triage data currently exists only on paper, and even there, only the front page (ESI, vitals) is reliably completed. Before any classifier — rule-based or ML — can be trained or validated, structured digital data must exist at the moment of capture, not as a later transcription step prone to loss and delay. This is the foundational justification for the entire Phase 1 proposal.

Constraint 3 — Shared Investigations Bottleneck

Labs and radiology are shared hospital-wide resources with 30-minute to 3-hour turnaround, and ED workups compete directly with elective inpatient demand. This is the dominant flow bottleneck at Mercer. Any AI design that increases test-ordering volume without addressing turnaround capacity will worsen overcrowding rather than relieve it — this constraint actively rules out certain plug-in points (see Investigations stage in diagram above) rather than just shaping their design.

Clinical Stakeholders

The **five identified clinical stakeholders** at Mercer General, each with a distinct top concern that any triage tool should respect are:

Stakeholder	**Role Category**	**Top Concern**
Triage Nurse	Nurse	They make the right ESI call quickly, and attend to patients in the waiting room who have just gotten worse.
ED Physician / Consultant on Call	Consultant	They handle the workload and documentation burden. They need decision support that doesn't slow them down or add to charting time.
EMS Crew	Porter / Transport	They need a clean, SBAR-style handover. Additional waiting hours to clear their stretcher for the next call impede them.
Registration Clerk	Admin	They need to get demographics right at a moment when the patient may be frightened, unwell, or unable to communicate clearly.

Patient & Family	Patient Advocate	They need to be informed of what is happening, Especially when activity in the department appears slow from the outside.

Do We Consider the Porter Role and Do We Consider It?

From the reading, Mercer General does not have a dedicated porter position documented in the available operational brief. Ergo, EMS crews are used here as the closest functional equivalent. They're responsible for a physical handoff into the department, with a comparable concern around minimising delay at that handoff point.

Proposed Solution

This project proposes a two-phase approach.

1. **Phase 1**, the 12-week pilot, focuses on designing and validating a structured digital triage capture tool aligned with the existing Mercer General paper triage form, producing a clean, de-identified, queryable dataset from real ED encounters.
2. **Phase 2**, beyond this programme, would use the developed dataset to train and locally validate a rule-based ESI classifier, iterating toward a machine learning model as data volume grows.

The Phase 1 deliverable is not an AI model. It is the dataset that makes a future model possible — and as Dr De Fretias has noted, this alone is genuinely valuable.

The tool would be designed for the nurse-led triage model at Mercer, with a minimal friction interface that respects the 3–5 minute triage window and does not increase documentation burden.

Conflicts, Stakeholders, and Workflow, and How They Apply to the Solution.

This week's systems-thinking analysis strengthens the proposal in two concrete ways.

1. First, the workflow diagram makes explicit where the Phase 1 tool sits—the Registration/Triage handoff, and where future Phase 2 decision support would sit—the zone-placement decision point. These are no longer abstract phases but specific, located interventions.

2. Second, the constraints and stakeholder analysis converts Sister Patrice's scepticism into design requirements the tool must:

    a. Respect the friction budget (Constraint 1).

    b. Capture data at the point of care rather than after (Constraint 2).

    c. Avoid worsening the one bottleneck that already defines ED flow (Constraint 3).

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Glossary

A reference index for clinical, technical, and research terminology used throughout this proposal. Intended for any reviewer — clinical or non-clinical — who needs a quick definition without breaking flow.

Clinical & Vital Sign Terms

Term	Meaning
ED	Emergency Department
ESI	Emergency Severity Index — the 5-level triage scale used at Mercer General (1 = immediate, 5 = non-urgent)
SBP	Systolic Blood Pressure — the peak pressure during a heartbeat (top number in a BP reading)
DBP	Diastolic Blood Pressure — the pressure between heartbeats (bottom number in a BP reading)

MAP	Mean Arterial Pressure — average pressure driving blood to organs across a cardiac cycle; calculated as (SBP + 2×DBP) / 3
GCS	Glasgow Coma Scale — score from 3–15, measuring the level of consciousness
RR	Respiratory Rate — breaths per minute
FiO2	Fraction of Inspired Oxygen — the percentage of oxygen a patient is receiving (room air = 21%)
SpO2	Oxygen Saturation — percentage of haemoglobin carrying oxygen, measured via pulse oximeter
CTAS	Canadian Triage and Acuity Scale — a triage system similar to ESI, used as a comparison benchmark in some studies
SBAR	Situation, Background, Assessment, Recommendation — a structured format for clinical handover communication
LOS	Length of Stay — total time a patient spends in the ED, from arrival to exit

DNR / DNI	Do Not Resuscitate / Do Not Intubate — patient code status designations

Healthcare Systems & Infrastructure Terms

Term	Meaning
EHR	Electronic Health Record — digital patient record system; Mercer currently has none for triage, hence the Phase 1 proposal
LMIC	Low- and Middle-Income Country — a World Bank income classification used in global health literature
SIDS	Small Island Developing State — a UN classification for small island nations (e.g. Vanuatu, many Caribbean states) facing shared infrastructure constraints
Boarding time	The gap between a disposition decision (e.g. "admit") and the patient physically leaving the ED, usually due to no ward bed being available

AI / Machine Learning Terms

Term	Meaning
AI	Artificial Intelligence
ML	Machine Learning
AUROC	Area Under the Receiver Operating Characteristic curve — a model performance metric from 0.5 (no better than chance) to 1.0 (perfect); higher is better
XGBoost	Extreme Gradient Boosting — a machine learning algorithm that builds an ensemble of decision trees, commonly used for structured/tabular data like vitals
Rule-based classifier	A decision-making system using explicit if/then logic (like the Day 6 triage pseudocode) rather than a trained statistical model
Multimodal	Combining more than one data type (e.g. structured vitals + unstructured free-text notes) in a single model

Research & Documentation Terms

Term	Meaning
DOI	Digital Object Identifier — a permanent link format for academic papers (e.g. https://doi.org/...)
APA	American Psychological Association — the citation style used throughout this proposal (7th edition)
Scoping review	A type of literature review that maps the breadth of existing research on a topic, without necessarily assessing study quality in depth
Systematic review	A more rigorous literature review following a defined search and inclusion protocol is often used to assess evidence quality
Preprint	A paper shared publicly before completing formal peer review (e.g. via ResearchSquare)

Version Control & Engineering Terms

Term	Meaning

PR	Pull Request — a proposed set of code changes submitted for review before merging into the main branch
Feature branch	A separate copy of a codebase is used to make changes safely without affecting the main version
Mermaid	A text-based diagramming syntax that renders automatically as a flowchart/diagram when viewed on GitHub
.gitignore	A file listing what should NOT be tracked by version control (e.g. patient data, local environments)